

# Statistical Machine Learning

## Logistic Regression and SVM

Horacio Gómez-Acevedo  
Department of Biomedical Informatics

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# Logistic Regression

Consider that we have an experiment with covariate(s)  $X$  and response  $Y$  with either two possibilities *yes* or *no*. For convenience, we use the binary codes for the response variable, say 1 for *yes* and 0 for *no*. We use **logistic regression** to model the probability that  $Y$  is 1 or 0.

We use the following notation to denote the probability that given

$$p(X = x) = Pr(Y = 1|X = x)$$

## Logistic regression and classification

Logistic regression provides a probability model, using it for classification means that you know something about the outcome beyond your data.

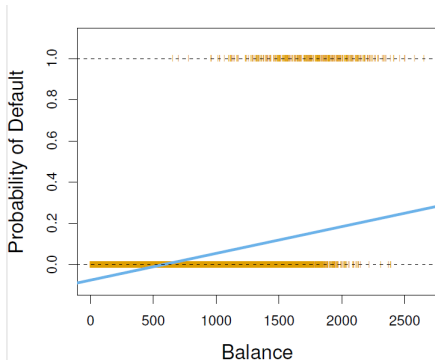
Thus, if  $p(X = x) > 0.5$  we can say that  $X = x$  belongs to the class 1. However, the cutoff used is something that requires justification.

## Logistic Regression (cont)

Our first attempt may be to use linear regression as follows:

$$p(X) = \theta_0 + \theta_1 X$$

However, when we use such a model we may encounter negative probabilities!



# Logistic Function

We apply the so-called **logistic function**

$$p(X) = \frac{\exp(\theta_0 + \theta_1 X)}{1 + \exp(\theta_0 + \theta_1 X)}$$

After some algebraic manipulation, we can show that

$$\log \left( \frac{p(X)}{1 - p(X)} \right) = \theta_0 + \theta_1 X \quad (1)$$

The left-hand of (1) is called the **log-odds** or **logit**.

For prediction, in the logistic regression model we use  $\hat{p}(X)$  as follows

$$\hat{Y} = \begin{cases} 0 & \text{if } \hat{p}(X) \leq 0.5 \\ 1 & \text{if } \hat{p}(X) > 0.5 \end{cases}$$

# Logistic Regression Coefficients Interpretation

In the normal linear regression model  $\theta_1$  give the average change in  $Y$  associated with a one-unit increase in  $X$ . Say for instance we have a linear model

$$\text{sales} = \theta_0 + \theta_1(\text{TV units sold})$$

Thus, the  $E(\text{sales})$  when the TV units sold is increased by 1 unit is  $\theta_1$ . In a logistic regression model increasing  $X$  by one unit changes the log odds by  $\theta_1$ , or equivalently it multiplies the odds by  $\exp(\theta_1)$ .

# Estimating the coefficients

Whereas we can use our training data to estimate the coefficients  $\theta_0$  and  $\theta_1$  using a least squares approach, this is not used in logistic regression. Instead the method of **maximum likelihood** is applied.

The likelihood function is defined as the joint probability of the observed data values, given the model parameters. Formally, it is defined as

$$\ell(\theta_0, \theta_1) = \prod_{i: y_i=0} p(x_i) \prod_{i': y'_i=1} (1 - p(x_{i'}))$$

The **deviance** is defined as

$$\text{deviance} = -2 \log \text{likelihood}$$

The estimates  $\hat{\theta}_0$  and  $\hat{\theta}_1$  are chosen to *maximize* this likelihood function. Equivalently, we can *minimize* the deviance.

# Estimating the coefficients (cont)

The terminology for logistic regression and multiple linear regression is sometimes confusing, but they are describing the same basic principles.

| Regression                         | Logistic Regression                |
|------------------------------------|------------------------------------|
| Degrees of Freedom                 | Degrees of Freedom                 |
| Sum of squares (SS)                | Deviance (D)                       |
| Mean sum of squares (divide by DF) | Mean deviance (divide by DF)       |
| Fitting by minimizing residual SS  | Fitting by minimizing the deviance |

# Multiple Logistic Regression

As with multi linear regression, we can expand the model to accommodate more predictors

$$\log \left( \frac{p(X)}{1 - p(X)} \right) = \theta_0 + \theta_1 X_1 + \cdots + \theta_p X_p,$$

where  $X = (X_1, \dots, X_p)$  are  $p$  predictors. And the corresponding logistic function is given by

$$p(X) = \frac{\exp(\theta_0 + \theta_1 X_1 + \cdots + \theta_p X_p)}{1 + \exp(\theta_0 + \theta_1 X_1 + \cdots + \theta_p X_p)}$$



# Maximal Margin Classifiers

A **hyperplane of dimension  $p$**  consists of the points (vectors) in  $\mathbb{R}^p$  that satisfy the condition

$$\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = 0, \quad (2)$$

where  $X = (X_1, \dots, X_p)^t$  is an element of  $\mathbb{R}^p$ .

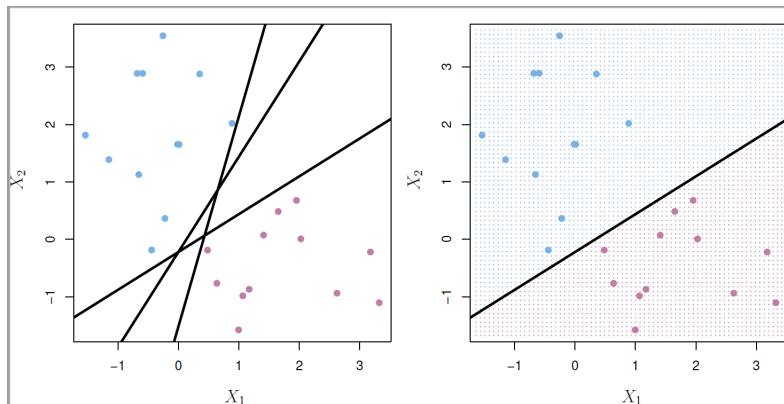
The hyperplane gives us a natural separation of  $\mathbb{R}^p$  into two regions

$$A^- = \{X \in \mathbb{R}^p : \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p < 0\}$$

$$A^+ = \{X \in \mathbb{R}^p : \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p > 0\}$$

# Separating Hyperplanes (Visual Idea)

The problem of separating by a hyperplane (line in  $\mathbb{R}^2$ , plane in  $\mathbb{R}^3$ , and hyperplane for higher dimensions)



# Classification by separating hyperplanes

Suppose we have a  $n \times p$  data matrix  $\mathbf{X}$  that consists of  $n$  training observations in the  $p$ -dimensional space

$$\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \dots, \mathbf{x}_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

and that these observations belong into either of the classes  $w_1$  or  $w_2$ . For simplicity sake, we label the classes as a form of a vector  $\mathbf{Y} = (y_1, \dots, y_p)$  where  $y_i \in \{-1, 1\}$  for each  $i$ .

# Classification by separating hyperplanes

If there is a hyperplane that separates the training observations perfectly according to their labels, we call it **separating hyperplane**. Then it satisfies the conditions

$$\begin{aligned}\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} &< 0 & \text{if } y_i = -1 \\ \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} &> 0 & \text{if } y_i = 1\end{aligned}$$

Alternatively, a separating hyperplane satisfies the condition

$$y_i (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}) > 0 \quad \text{for all } i \in \{1, \dots, n\}$$

# Maximal Margin Classifier

Normally, if we have a separating hyperplane, there are an infinite number of alternative hyperplanes. The question is how do we select one?

We will select an "optimal" plane by performing the following optimization process on the training data

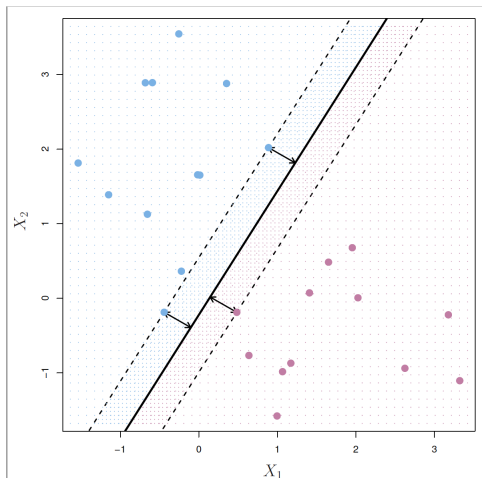
- Compute the (perpendicular) distance from each training point to the given separating hyperplane.
- The minimum distance from the hyperplane defines the **margin**.
- Select the hyperplane with the maximal margin  $M$ .

The classification is carried out in test data by using the **maximal margin classifier**. That is the sign of  $f(x^*) = \beta_0 + \beta_1 x_1^* + \dots + \beta_p x_p^*$ .

The points that are equidistant from the maximal margin hyperplane are refer to as **support vectors**.

# Support Vectors

An important property is that the maximal margin hyperplane depends directly on only a small subset of the observations (the support vectors).



# Construction of the Maximal Margin Classifier

Finding the maximal margin hyperplane corresponds to finding a solution to the optimization problem

$$\begin{aligned} & \text{maximize}_{\beta_0, \dots, \beta_p} M \\ & \text{subject to } \sum_{j=1}^p \beta_j^2 = 1, \\ & y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq M \quad \forall i \in \{1, \dots, n\} \end{aligned}$$

where our  $n$  training observations are  $x_i$  and  $Y$  is the vector with the class labels.

## Non-separable Case

There are instances in which we cannot find the separating plane. One way to get around this problem is to introduce **soft margins**, which means that we allow observations to be on the incorrect side of the margin, or even the incorrect side of the hyperplane.

$$\begin{aligned} & \text{maximize } \beta_0, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n M \\ & \text{subject to } \sum_{j=1}^p \beta_j^2 = 1, \\ & y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq M(1 - \epsilon_i) \\ & \epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C \end{aligned}$$

where  $C$  is a hyperparameter that will be tuned.



# Support Vector Classifier

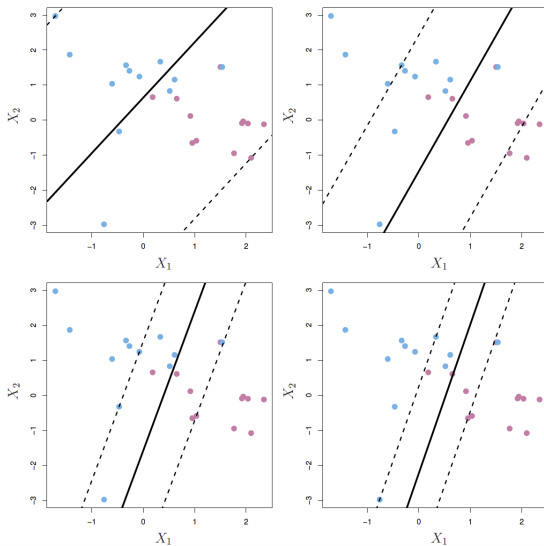
The variables  $\epsilon_i$  are called **slack variables** that allow individual observations to be on the wrong side of the margin or the hyperplane. More precisely, if  $\epsilon_i = 0$  the  $i$ th observation is on the right side of the margin. If  $0 < \epsilon_i \leq 1$ , the  $i$ th observation is on the wrong side of the margin. If  $\epsilon_i > 1$  the observation  $i$ th is on the wrong side of the hyperplane.

The hyperparameter  $C$  limits the tolerance that we allow for the observations to be either on the wrong side of the margin or hyperplane. It controls the bias-variance trade-off.

## Support Vector Classifier (cont)

- If  $C$  is large, the margin is wider and we allow more violations; this amounts to fitting the data less hard and increasing bias but decreasing variance.
- If  $C$  is small, we fit very tightly our model, so a low bias but high variance.

# Variable C



# Support Vector Machines

We generalize the previous problem by expanding the number of features. Say, instead of fitting a support vector classifier using  $p$  features

$$X_1, \dots, X_p,$$

we could instead fit a support vector classifier using  $2p$  features

$$X_1, X_1^2, X_2, X_2^2, \dots, X_p, X_p^2$$

In this case, the optimization process becomes

$$\begin{aligned} & \text{maximize } \beta_0, \beta_{11}, \beta_{12}, \dots, \beta_{1p}, \beta_{2p}, \epsilon_1, \dots, \epsilon_n M \\ & \text{subject to } y_i \left( \beta_0 + \sum_{j=1}^p \beta_{j1} x_{ij} + \sum_{j=1}^n \beta_{j2} x_{ij}^2 \right) \geq M(1 - \epsilon_i) \\ & \sum_{j=1}^p \sum_{j=1}^2 \beta_{ij}^2 = 1, \epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C \end{aligned}$$

# Support Vector Machines

As we saw in the example above, one way to extend the support vector classifier is enlarging the feature space by the so-called **kernels**. Instead of using the observations, we will use the **inner product** of them. The dot product is a form of an inner product

$$\langle x_i, x_{i'} \rangle = \sum_{j=1}^p x_{ij} x_{i'j}$$

In this framework, the linear support classifier can be represented as

$$f(x) = \beta_0 + \sum_{i=1}^n \alpha_i \langle x, x_i \rangle$$

Note that for the estimation of  $\beta_0$  and  $\alpha_i$  we need the  $\binom{n}{2}$  inner products  $\langle x_i, x_{i'} \rangle$  between all pairs of training observations. However, this expression simplifies because if  $x_i$  is not a support vector  $\alpha_i = 0$ .

## Other kernels

We can replace the inner product with other kernels:

- Polynomial kernel of degree  $d$

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^p x_{ij} x_{i'j}\right)^d$$

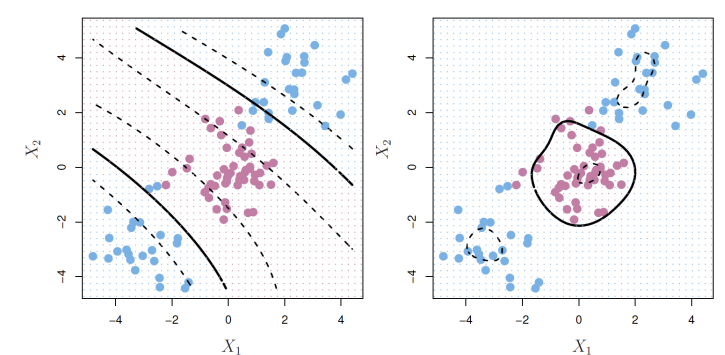
- Radial kernel

$$K(x_i, x_{i'}) = \exp \left( -\gamma \sum_{j=1}^p (x_{ij} - x_{i'j})^2 \right)$$

- Neural network

$$K(x_i, x_{i'}) = \tanh(\kappa_1 \langle x, x' \rangle + \kappa_2)$$

# SVMs with different kernels



In the left, we have a polynomial kernel of degree 3, on the right we have a radial kernel.

# Rough Optimization Ideas

Let's start with an equivalence

$$\begin{aligned} & \text{maximize } \beta_0, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n M \\ & \text{subject to } \sum_{j=1}^p \beta_j^2 = 1, \\ & y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq M(1 - \epsilon_i) \\ & \epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C \end{aligned}$$

With the following (dropping the norm constrain in  $\beta$  and define  $C = 1/\|\beta\|$ )

$$\begin{aligned} & \text{minimize } \beta_0, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n \|\beta\| \\ & \text{subject to } y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq (1 - \epsilon_i) \\ & \epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq \text{constant} \end{aligned} \tag{3}$$



# Optimization Problem

The problem (3) is quadratic with linear inequality constraints, hence it is a convex optimization problem. We rewrite this problem as

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 + \gamma \sum_{i=1}^N \epsilon_i$$

subject to  $\epsilon_i \geq 0, y_i(x_i^T \beta + \beta_0) \geq 1 - \epsilon_i \forall i$

The Lagrange (primal) function is

$$L_p = \frac{1}{2} \|\beta\|^2 + \gamma \sum_{i=1}^N \epsilon_i - \sum_{i=1}^N \alpha_i [y_i(x_i^T \beta + \beta_0) - (1 - \epsilon_i)] - \sum_{i=1}^N \mu_i \epsilon_i \quad (4)$$

which is minimized with respect to  $\beta, \beta_0$  and  $\epsilon_i$ . Using calculus we take partial derivatives and make them zero.

# Optimization Problem

$$\begin{aligned}\beta_j &= \sum_{i=1}^N \alpha_i y_i x_{ij} \\ 0 &= \sum_{i=1}^N \alpha_i y_i, \\ \alpha_i \gamma - \mu_i &\quad \forall i\end{aligned}\tag{5}$$

as well as the positivity constraints  $\alpha_i, \mu_i, \epsilon_i \geq 0$  for all  $i$ . If we substitute (5) into (4), we obtain the Lagrangian(Wolfe) dual objective function

$$L_D = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{i'=1}^N \alpha_i \alpha_{i'} y_i y_{i'} x_i^T x_{i'},\tag{6}$$

We maximize  $L_D$  subject to  $0 \leq \alpha_i \leq \gamma$  and  $\sum_{i=1}^N \alpha_i y_i = 0$ .

# What we have learned?

- Logistic regression as a classification process
- Maximal margin classifier as a method for classification
- Support Vector classifier uses a hyperparameter to control soft limits
- Kernels allow us to generalize the SVM to take care of the non-linearity.

# References

Materials and some of the pictures are from [1], [3], [2], [4] and [5].

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